

## Scalability & Future Plan

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|----------------------|----------------------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                                  |
| <b>Team ID</b>       | SWTID-2026-2508                                                |
| <b>Project Name</b>  | Streamlining IT Procurement: Automating Standard Laptop Orders |
| <b>Maximum Marks</b> | 1 Mark                                                         |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| <b>S.No</b> | <b>Limitation</b>                               | <b>Impact</b>                                                         | <b>Priority to Address (High / Medium / Low)</b> |
|-------------|-------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------|
| 1           | Orders are not sent to suppliers automatically. | Procurement must place each order by hand, which adds time.           | High                                             |
| 2           | Supports standard laptop models only.           | Other IT hardware cannot be requested through the system.             | Medium                                           |
| 3           | Single-level approval.                          | Organizations that need several approvers cannot model their process. | Medium                                           |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                   | Proposed Upgrade / Solution                                                                       |
|------|--------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1    | User Load          | One Flask app instance, suited to a small team. | Run several Gunicorn workers behind an Nginx load balancer and host on a cloud server.            |
| 2    | Data Storage       | One MySQL database on a single machine.         | Regular backups, indexes on request and order tables, and a managed cloud MySQL with replication. |
| 3    | Performance        | Every page reads directly from the database.    | Paginate long request tables and cache the catalog and dashboard counts.                          |
| 4    | Security           | Hashed passwords and role-based access.         | Add HTTPS, stronger password rules, audit logs of approvals and optional single sign-on.          |

### Future Roadmap:

| Phase   | Planned Feature / Enhancement                                         | Target Timeline  | Expected Impact                                            |
|---------|-----------------------------------------------------------------------|------------------|------------------------------------------------------------|
| Phase 2 | Supplier API integration so approved orders are placed automatically. | Within 3 months  | Removes manual ordering and speeds up delivery.            |
| Phase 3 | Multi-level approvals and support for other IT hardware.              | Within 6 months  | Fits larger organizations and more types of requests.      |
| Phase 4 | Cloud hosting, mobile-friendly app and advanced analytics.            | Within 12 months | Supports more users and gives better procurement insights. |